CSE 270 Sports Analytics

Predicting with models

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Poisson regression

Predicting game outcome with Poisson regression

Poisson regression

First we are going to test the hypothesis that Home goals and Away goals are independent.

We will use a χ^2 test of independence of two categorical variables

$$H_0 : \text{Two variables are independent}$$

$$H_1 : \text{Two variables are NOT independent}$$

Calculate p-value and compare with significance level of α, if p-value $< \alpha$ reject the NULL hypothesis

Poisson regression

First we are going to create categoric variables for Home and Away team goals in a way that more than 3 goals will be combined together

```
library(dplyr)
data("f_data_sm")
df <- f_data_sm %>% mutate(FTAG = ifelse(FTAG>=4, "4+", FTAG),
                           FTHG = ifelse(FTHG >=4, "4+", FTHG))
```

Poisson regression

Recall that χ^2 test is using Expected and Observed frequencies as an input. Thus we will need to provide a table with joint frequencies of Home and away team goals

```
table(Away=df$FTAG, Home = df$FTHG)
```

```
##      Home
## Away     0     1     2     3    4+
##    0  7059  9325  7261  3998  2756
##    1  6074 10248  7644  4020  2621
##    2  3858  5243  4277  2092  1284
##    3  1688  2313  1464   826   462
##    4+ 1002  1103   742   300   154
```

Poisson regression

We do the test for the last season

- We use function chisq.test
- We specify table with Observed Frequencies
- And specify simulate.p.values = TRUE: This will add robustness to our test
- We do the test for each country and extract p-values of the test

Poisson regression

```
df %>% filter(SEASON==2019) %>% group_by(COUNTRY) %>%
  summarise(pvalue=chisq.test(table(FTAG, FTHG),
                              simulate.p.value=T)$p.value)
```

```
## # A tibble: 11 x 2
##    COUNTRY      pvalue
##    <chr>         <dbl>
##  1 Belgium     0.239
##  2 England     0.00300
##  3 France      0.115
##  4 Germany     0.106
##  5 Greece      0.100
##  6 Italy       0.159
##  7 Netherlands 0.0275
##  8 Portugal    0.334
##  9 Scotland    0.205
## 10 Spain       0.846
## 11 Turkey      0.294
```

The hypothesis of independence is not rejected, at $\alpha = 0.05$, Thus they are independent

Poisson regression

- Poisson regression is used when the dependent variable is a count data, i.e. y=0,1,2,3. . . , thus we assume that the dependent variable has Poisson distribution
- Independent variables can be both numeric and categorical
- Poisson regression is a part of Generalized Linear Regression (GLM) models family

Poisson regression: the general definition

In this chapter we are fitting the model of the following form

$$log(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ...$$

Where λ is the mean of Poisson distribution and $X_1, ..X_n$ are the independent variables

With the estimate of the λ we can generate Poisson distribution and calculate the probabilities for the number of the goals

Preparing data for the Poisson regression model

The initial data for Portugal, season 2018-2019

Making Predictions for Portugal Liga 1

```
data(f_data_sm)
portugal <- f_data_sm %>%
            filter(SEASON==2019, COUNTRY=="Portugal") %>%
            select(HOMETEAM, AWAYTEAM, FTHG, FTAG)

head(portugal, n=5)
```

```
##   HOMETEAM   AWAYTEAM FTHG FTAG
## 1  Benfica  Guimaraes    3    2
## 2    Porto     Chaves    5    0
## 3  Setubal       Aves    2    0
## 4  Tondela Belenenses    0    1
## 5 Feirense    Rio Ave    2    0
```

```
dim(portugal)
```

```
## [1] 306   4
```

Preparing data for the Poisson regression model

The data need to be in the following format: 2 rows for each game and the indicator showing if the team is a home team

```
port1<-data.frame(portugal[,c("HOMETEAM", "AWAYTEAM", "FTHG")], Home=1)
port2<-data.frame(portugal[,c("AWAYTEAM", "HOMETEAM", "FTAG")], Home=0)
colnames(port1)<-c("Team", "Opponent", "Goal", "Home")
colnames(port2)<-c("Team", "Opponent","Goal", "Home")
portugal2<-rbind(port1, port2)
head(portugal2)
```

```
##       Team    Opponent Goal Home
## 1  Benfica   Guimaraes    3    1
## 2    Porto      Chaves    5    1
## 3  Setubal        Aves    2    1
## 4  Tondela  Belenenses    0    1
## 5 Feirense     Rio Ave    2    1
## 6 Maritimo Santa Clara    1    1
```

Preparing data for the Poisson regression model

Now lets have a look at the first game of the season: Benfica vs Guimaraes * Benfica was the home team, and the game ended with 3-2

```
portugal[1,]
```

```
##   HOMETEAM  AWAYTEAM FTHG FTAG
## 1  Benfica Guimaraes    3    2
```

- Lets look at our new dataset

```
portugal2[1,]
```

```
##      Team  Opponent Goal Home
## 1 Benfica Guimaraes    3    1
```

```
portugal2[307,]
```

```
##          Team Opponent Goal Home
## 307 Guimaraes  Benfica    2    0
```

The number of the rows in the portugal2 dataset is twice of the original one

```
dim(portugal)
```

```
## [1] 306   4
```

```
dim(portugal2)
```

```
## [1] 612   4
```

The model

```
model<-glm(Goal~Team+Opponent+Home, data=portugal2, family=poisson(link=log))
coefficients(model)
```

```
##          (Intercept)       TeamBelenenses          TeamBenfica
##          -0.04950176           0.18546812           1.06027038
##         TeamBoavista          TeamChaves         TeamFeirense
##          -0.04047050          -0.01881011          -0.24101272
##        TeamGuimaraes         TeamMaritimo       TeamMoreirense
##           0.25493167          -0.30424065           0.10213559
##         TeamNacional     TeamPortimonense           TeamPorto
##          -0.02792327           0.24259884           0.71365399
##          TeamRio Ave      TeamSanta Clara          TeamSetubal
##           0.36181966           0.20134445          -0.23625112
##         TeamSp Braga        TeamSp Lisbon          TeamTondela
##           0.45606737           0.70311363           0.14038603
##   OpponentBelenenses      OpponentBenfica     OpponentBoavista
##           0.04912813          -0.36998629          -0.20470680
##       OpponentChaves     OpponentFeirense    OpponentGuimaraes
##           0.15040090           0.25756387          -0.35243515
##     OpponentMaritimo   OpponentMoreirense     OpponentNacional
##          -0.11925272          -0.10282703           0.39741274
## OpponentPortimonense       OpponentPorto      OpponentRio Ave
##           0.19802795          -0.84856604           0.07904179
##  OpponentSanta Clara      OpponentSetubal     OpponentSp Braga
##          -0.07516828          -0.23759199          -0.25481347
##    OpponentSp Lisbon      OpponentTondela                 Home
##          -0.34879055           0.10390298           0.25808160
```

Making predictions

- Benfica is playing against Porto at home

```
options(scipen=1, digits=4)
predict(model, data.frame(Home=1, Team="Benfica", Opponent="Porto"), type="response"
```

```
##     1
## 1.522
```

- The same for Porto

```
options(scipen=1, digits=4)
predict(model, data.frame(Home=0, Team="Porto", Opponent="Benfica"),
        type="response")
```

```
##     1
## 1.342
```

When Benfica is playing Porto at Home, the expected (average) number of goals for Benfica is 1.522 and For Porto 1.342

Lets do the calculations by hand for Benfica

Calculations of λ for Benfica

$$log(\lambda) = \beta_0 + \beta_{TeamBenfica} + \beta_{OpponentPorto} + \beta_{Home}$$

```
options(scipen=1, digits=4)
coefficients(model)["(Intercept)"]
```

```
## (Intercept)
##     -0.0495
```

```
coefficients(model)["TeamBenfica"]
```

```
## TeamBenfica
##        1.06
```

```
coefficients(model)["OpponentPorto"]
```

```
## OpponentPorto
##       -0.8486
```

```
coefficients(model)["Home"]
```

```
##   Home
## 0.2581
```

$log(\lambda) = -0.0495 + 1.06 + (-0.8486) + 0.2581 \quad \lambda = exp(0.42) = 1.52$

Making predictions: Home advantage

The average expected goals of the home team divided by the average expected goals by the same team playing away will be equal to exponent of Home coefficient

```
# lambda for Benfica playing at home
b_h<-exp(-0.0495+1.06-0.8486 + 0.2581)
# lambda for Benfica playing against Porto away
b_a<-exp(-0.0495+1.06-0.8486)
b_h/b_a
```

```
## [1] 1.294
```

```
exp(coefficients(model)["Home"])
```

```
##  Home
## 1.294
```

Making predictions

As we have λ for both teams, we can use skellam distribution to come up with the probabilities.

```
# Benfica's lambda
1.522
```

```
## [1] 1.522
```

```
# Porto's lambda
1.342
```

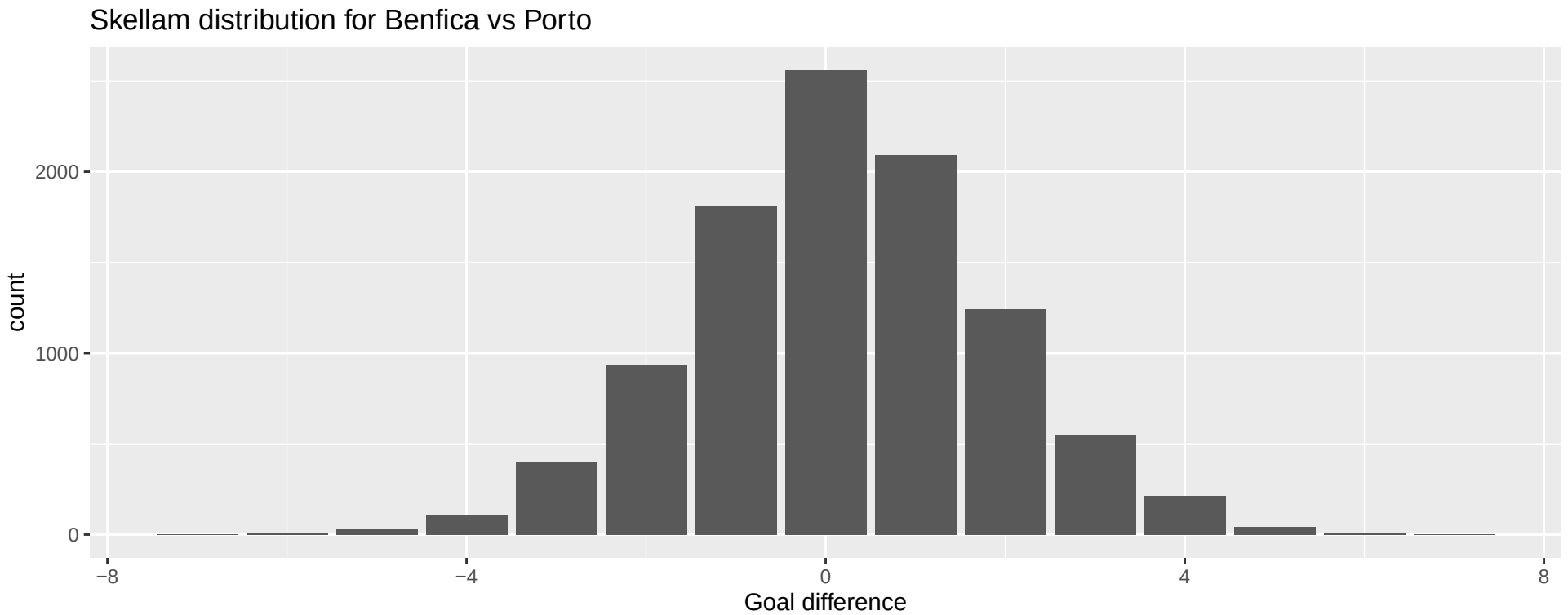
```
## [1] 1.342
```

```
# Simulate game results
set.seed(1)
k <- rskellam(10000,mu1 = 1.522, mu2 = 1.342)
```

Making predictions

A slight home team advantage

```
ggplot()+geom_bar(aes(x=k))+
  labs(x="Goal difference", title="Skellam distribution for Benfica vs Porto")
```



Making predictions

Calculate the game output probabilities

```
# Away team to win
sum(dskellam(c(-100:-1),1.522, 1.342))
```

```
## [1] 0.3354
```

```
# Draw
sum(dskellam(0 ,1.522, 1.342))
```

```
## [1] 0.2484
```

```
# Home team to win
sum(dskellam(c(1:100),1.522, 1.342))
```

```
## [1] 0.4162
```

Predicting with models

Bradley-Terry model

Bradley-Terry model

The Bradley-Terry model (Bradley and Terry 1952) assumes that in a 'contest' between any two 'players', say player i and player j ($i, j \in \{1, \ldots . , K\}$) the ,odds that i beats j are α_i/α_j , where α_i and α_j are positive-valued parameters which represent the "ability" of each player/team.

The general model definition

$$P(\text{i beats j}) = \frac{\alpha_i}{\alpha_i+\alpha_j}$$

Sometimes the modification with exponential score function

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i}+e^{\alpha_j}}$$

Bradley-Terry model

$P(\text{i beats j}) = \frac{\alpha_i}{\alpha_i+\alpha_j}$

$logit(P(\text{i beats j})) = \log(\frac{P(\text{i beats j})}{P(\text{j beats i})}) = \log(\frac{P(\text{i beats j})}{1-P(\text{i beats j})}) = log(\frac{\alpha_i}{\alpha_i+\alpha_j} * \frac{\alpha_i+\alpha_j}{\alpha_j}) =$
$log(\alpha_i) - log(\alpha_j) = \lambda_i - \lambda_j$

The goal of the modeling is to find the best estimate for λ_i, λ_j

Bradley-Terry model

The modification with exponential score function

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

$$logit(P(\text{i beats j})) = \log(\frac{P(\text{i beats j})}{P(\text{j beats i})}) = \log(\frac{P(\text{i beats j})}{1 - P(\text{i beats j})}) = log(\frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}} * \frac{e^{\alpha_i} + e^{\alpha_j}}{e^{\alpha_j}}) = \alpha_i - \alpha_j$$

logit - log of odds

Bradley-Terry model

The basic Bradley-Terry model does not predict draw/tie games. You can use Davidson's extension on Bradley-Terry model if want to predict draw:
https://www.jstor.org/stable/2283595?seq=1#page_scan_tab_contents

Bradley-Terry model

```
library(SportsAnalytics270)
data(nfl)
```

```
head(nfl, n=5)
```

```
##   schedule_season schedule_week schedule_date           team_away score_away
## 1            2000             1    2000-09-03  San Francisco 49ers         28
## 2            2000             1    2000-09-03     Tennessee Titans         13
## 3            2000             1    2000-09-03 Jacksonville Jaguars         27
## 4            2000             1    2000-09-03  Philadelphia Eagles         41
## 5            2000             1    2000-09-03        New York Jets         20
##           team_home score_home              stadium stadium_type
## 1   Atlanta Falcons         36         Georgia Dome       indoor
## 2     Buffalo Bills         16 Ralph Wilson Stadium      outdoor
## 3  Cleveland Browns          7  FirstEnergy Stadium      outdoor
## 4    Dallas Cowboys         14        Texas Stadium      outdoor
## 5 Green Bay Packers         16        Lambeau Field      outdoor
##   stadium_location result spread home_odds_open away_odds_open home_odds_close
## 1      Atlanta, GA      H     -8             NA             NA              NA
## 2 Orchard Park, NY      H     -3             NA             NA              NA
## 3    Cleveland, OH      A     20             NA             NA              NA
## 4       Irving, TX      A     27             NA             NA              NA
## 5    Green Bay, WI      A      4             NA             NA              NA
##   away_odds_close
## 1              NA
## 2              NA
## 3              NA
## 4              NA
## 5              NA
```

Bradley-Terry model

- Data on NFL matches
- Description
- Format
- A data frame with 4848 rows and 16 variables
- schedule_season: from 2000-2016
- schedule_week: The number of the week when the game was played
- schedule_date: The actual date of the game
- team_away: Away team
- score_away: The score of the away team
- team_home: Home team
- score_home: The score of home team
- stadium: The name of the stadium
- stadium_type: indoor, outdoor, retractable
- stadium_location: The location of the stadium: City and state abbreviation
- odds: closing and opening betting odds
- result: The result of the game: H=Home Win, T=Tie, A=Away Win
- spread: points spread, the difference between away team and home team scores ,
- home_odds_open, home_odds_close: Opening and closing decimal odds for the home team
- home_odds_open, away_odds_close: Opening and closing decimal odds for the away team

Bradley-Terry model

Some data preparation

```
nfl2 <- nfl%>%
  select(team_home, team_away, result) %>%
  filter(result !="T") %>%
  mutate(ht_w = ifelse(result=="H",1,0),
         at_w = ifelse(result=="A",1,0))
head(nfl2)
```

```
##            team_home            team_away result ht_w at_w
## 1    Atlanta Falcons  San Francisco 49ers      H    1    0
## 2      Buffalo Bills     Tennessee Titans      H    1    0
## 3   Cleveland Browns Jacksonville Jaguars      A    0    1
## 4     Dallas Cowboys  Philadelphia Eagles      A    0    1
## 5  Green Bay Packers        New York Jets      A    0    1
## 6 Kansas City Chiefs   Indianapolis Colts      A    0    1
```

Bradley-Terry model

The final dataset

```
library(dplyr)
nfl3 <- nfl2 %>%
    mutate(team_home=as.factor(team_home),
           team_away = as.factor(team_away)) %>%
    group_by(team_home, team_away) %>%
    summarise(ht=sum(ht_w), at=sum(at_w))

head(nfl3, n=5)
```

```
## # A tibble: 5 x 4
## # Groups:   team_home [1]
##   team_home          team_away            ht    at
##   <fct>              <fct>             <dbl> <dbl>
## 1 Arizona Cardinals Atlanta Falcons       2     1
## 2 Arizona Cardinals Baltimore Ravens      1     2
## 3 Arizona Cardinals Buffalo Bills         1     1
## 4 Arizona Cardinals Carolina Panthers     2     4
## 5 Arizona Cardinals Chicago Bears         0     3
```

Bradley-Terry model

Fitting the bradley-terry model

```
install.packages("BradleyTerry2")
```

```
library(BradleyTerry2)
model <- BTm(cbind(ht, at), team_home, team_away, data=nfl3, id="team_")
```

Bradley-Terry model

Coefficients represent the teams abilities to win

```
summary(model)
```

```
##
## Call:
## BTm(outcome = cbind(ht, at), player1 = team_home, player2 = team_away,
##     id = "team_", data = nfl3)
##
## Deviance Residuals:
##    Min      1Q  Median      3Q     Max
## -2.736  -0.270   0.282   1.201   3.251
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## team_Atlanta Falcons        0.3675     0.1645    2.23  0.02547 *
## team_Baltimore Ravens       0.6114     0.1699    3.60  0.00032 ***
## team_Buffalo Bills          0.0525     0.1699    0.31  0.75727
## team_Carolina Panthers      0.3060     0.1639    1.87  0.06190 .
## team_Chicago Bears          0.2415     0.1654    1.46  0.14414
## team_Cincinnati Bengals     0.2336     0.1696    1.38  0.16829
## team_Cleveland Browns      -0.4206     0.1744   -2.41  0.01590 *
## team_Dallas Cowboys         0.4127     0.1635    2.52  0.01161 *
## team_Denver Broncos         0.6187     0.1687    3.67  0.00024 ***
## team_Detroit Lions         -0.2257     0.1662   -1.36  0.17451
## team_Green Bay Packers      0.6829     0.1674    4.08  4.5e-05 ***
## team_Houston Texans         0.0891     0.1737    0.51  0.60807
## team_Indianapolis Colts     0.8033     0.1711    4.70  2.7e-06 ***
## team_Jacksonville Jaguars  -0.0820     0.1697   -0.48  0.62895
## team_Kansas City Chiefs     0.3332     0.1676    1.99  0.04683 *
## team_Los Angeles Chargers   0.3680     0.1676    2.20  0.02813 *
## team_Los Angeles Rams       0.0491     0.1566    0.31  0.75362
## team_Miami Dolphins         0.2876     0.1692    1.70  0.08914 .
## team_Minnesota Vikings      0.3260     0.1647    1.98  0.04779 *
## team_New England Patriots   1.3484     0.1798    7.50  6.4e-14 ***
## team_New Orleans Saints     0.5057     0.1655    3.06  0.00224 **
## team_New York Giants        0.3226     0.1635    1.97  0.04855 *
```

Bradley-Terry model

New England Patriots: the strongest team Cleveland Browns: the weakest team

```
coef<-model$coefficients
sort(coef, decreasing = T)
```

```
## team_New England Patriots  team_Pittsburgh Steelers   team_Indianapolis Colts
##                    1.34835                   0.90424                   0.80328
##    team_Green Bay Packers  team_Philadelphia Eagles       team_Denver Broncos
##                    0.68295                   0.65415                   0.61867
##     team_Baltimore Ravens   team_New Orleans Saints     team_Seattle Seahawks
##                    0.61137                   0.50572                   0.47580
##       team_Dallas Cowboys team_Los Angeles Chargers      team_Atlanta Falcons
##                    0.41274                   0.36800                   0.36746
##   team_Kansas City Chiefs    team_Minnesota Vikings      team_New York Giants
##                    0.33316                   0.32602                   0.32258
##    team_Carolina Panthers       team_Miami Dolphins     team_Tennessee Titans
##                    0.30602                   0.28758                   0.28242
##        team_Chicago Bears   team_Cincinnati Bengals       team_New York Jets
##                    0.24153                   0.23362                   0.22661
##       team_Houston Texans        team_Buffalo Bills  team_San Francisco 49ers
##                    0.08908                   0.05251                   0.05031
##     team_Los Angeles Rams team_Tampa Bay Buccaneers  team_Washington Redskins
##                    0.04914                   0.04106                   0.03768
## team_Jacksonville Jaguars      team_Oakland Raiders       team_Detroit Lions
##                   -0.08200                  -0.10695                  -0.22572
##     team_Cleveland Browns
##                   -0.42056
```

Bradley-Terry model

Overall there are 32 teams in the data

```
length(unique(nfl3$team_home))
```

```
## [1] 32
```

```
# and we have coefficients for 31
length(coef)
```

```
## [1] 31
```

- One team is taken as a reference level
- Its ability is $= 0$
- In our case that is Arizona Cardinals as it comes first alphabetically

Bradley-Terry model

- Abilities are calculated relative to Arizona Cardinales
- Thus, positive ability means that the team is doing better than Cardinals
- Negative ability is showing that the team is doing worst than Cardinals

```
BTabilities(model)
```

```
##                       ability   s.e.
## Arizona Cardinals     0.00000 0.0000
## Atlanta Falcons       0.36746 0.1645
## Baltimore Ravens      0.61137 0.1699
## Buffalo Bills         0.05251 0.1699
## Carolina Panthers     0.30602 0.1639
## Chicago Bears         0.24153 0.1654
## Cincinnati Bengals    0.23362 0.1696
## Cleveland Browns     -0.42056 0.1744
## Dallas Cowboys        0.41274 0.1635
## Denver Broncos        0.61867 0.1687
## Detroit Lions        -0.22572 0.1662
## Green Bay Packers     0.68295 0.1674
## Houston Texans        0.08908 0.1737
## Indianapolis Colts    0.80328 0.1711
## Jacksonville Jaguars -0.08200 0.1697
## Kansas City Chiefs    0.33316 0.1676
## Los Angeles Chargers  0.36800 0.1676
## Los Angeles Rams      0.04914 0.1566
```

Bradley-Terry model

- Let's do some plotting
- First let's create a dataframe with abilities

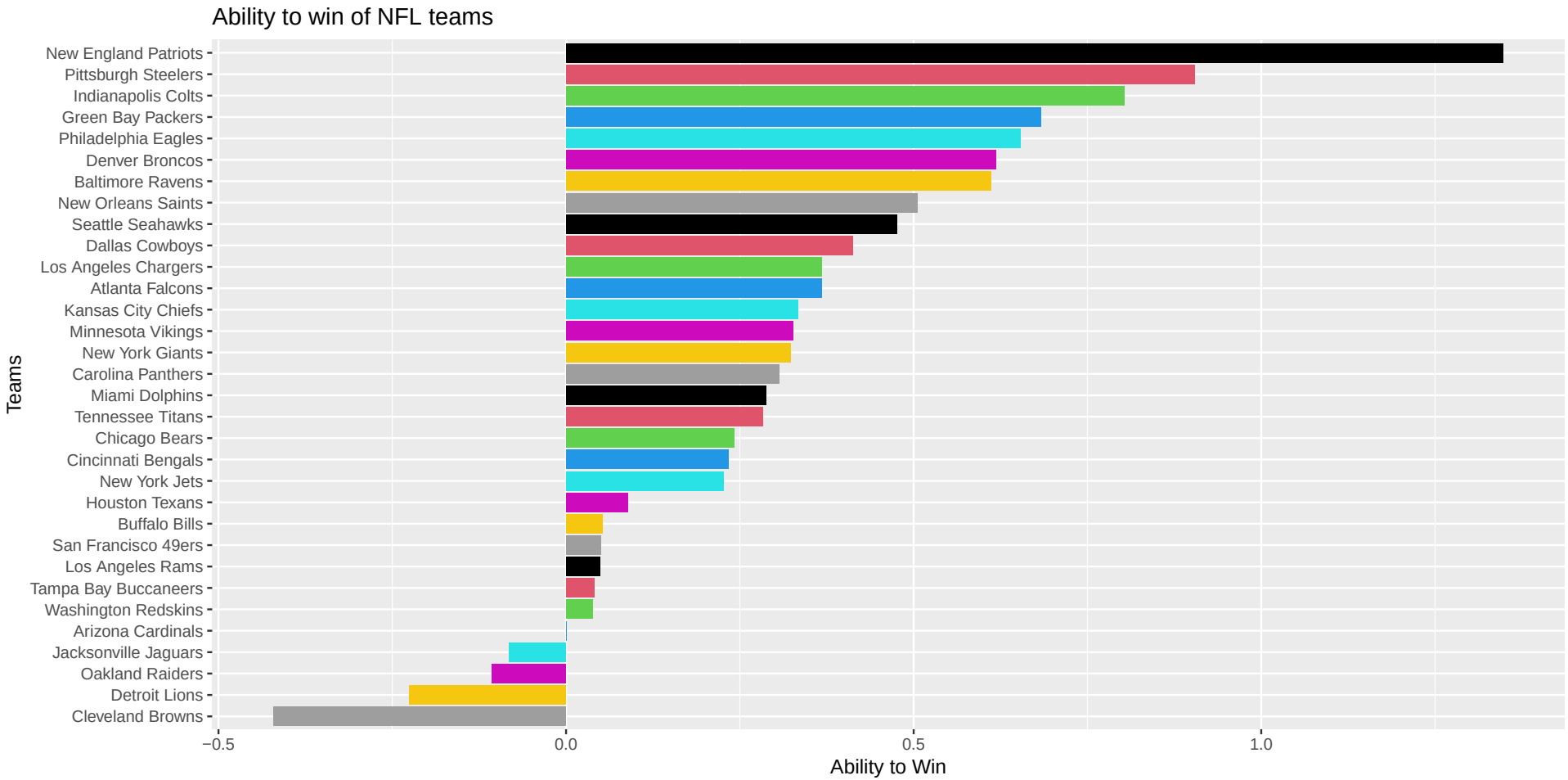
```
abilities <- as.data.frame(BTabilities(model))
abilities$team <- rownames(abilities)
abilities <- abilities[order(abilities$ability, decreasing = T),]
abilities
```

```
##                         ability   s.e.                    team
## New England Patriots    1.34835 0.1798 New England Patriots
## Pittsburgh Steelers     0.90424 0.1733  Pittsburgh Steelers
## Indianapolis Colts      0.80328 0.1711   Indianapolis Colts
## Green Bay Packers       0.68295 0.1674    Green Bay Packers
## Philadelphia Eagles     0.65415 0.1650  Philadelphia Eagles
## Denver Broncos          0.61867 0.1687       Denver Broncos
## Baltimore Ravens        0.61137 0.1699     Baltimore Ravens
## New Orleans Saints      0.50572 0.1655   New Orleans Saints
## Seattle Seahawks        0.47580 0.1574     Seattle Seahawks
## Dallas Cowboys          0.41274 0.1635       Dallas Cowboys
## Los Angeles Chargers    0.36800 0.1676 Los Angeles Chargers
## Atlanta Falcons         0.36746 0.1645      Atlanta Falcons
## Kansas City Chiefs      0.33316 0.1676   Kansas City Chiefs
## Minnesota Vikings       0.32602 0.1647    Minnesota Vikings
## New York Giants         0.32258 0.1635      New York Giants
## Carolina Panthers       0.30602 0.1639    Carolina Panthers
```

Bradley-Terry model

Abilities from strongest to weakest

```
ggplot(data=abilities, aes(x=reorder(team,ability), y=ability)) +
  geom_bar(stat="identity", fill=1:32) + coord_flip()+
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



Bradley-Terry model

everything relative to the worst team (Cleveland Browns)

- Lets say we want to calculate the abilities relative to the weakest team, so all the abilities would be ≥ 0
- You could do this by re leveling the team variables

```
nfl4<-nfl2 %>%
    mutate(team_home = relevel(team_home, ref="Detroit Lions"),
           team_away = relevel(team_away, ref="Detroit Lions")) %>%
    group_by(team_home, team_away) %>%
    summarise(ht=sum(ht_w), at=sum(at_w))

model_browns <- BTm(cbind(ht, at), team_home, team_away, data=nfl4, id="team_")
```

OR Standardize the variable ability around its minimum (these two approaches will give exactly the same result)

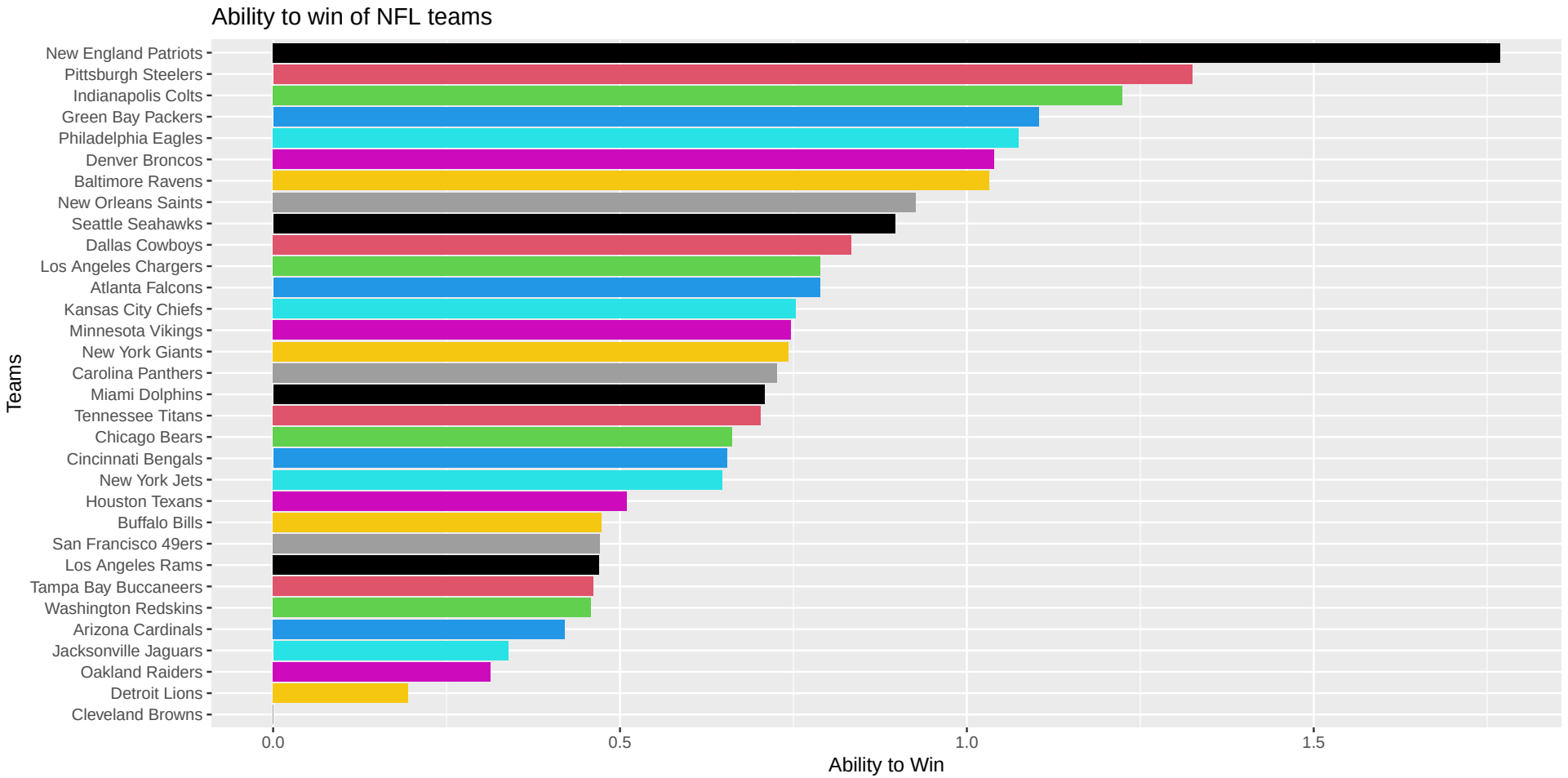
```
abilities_brown <- abilities
abilities_brown$ability <- abilities_brown$ability - min(abilities$ability)
head(abilities_brown)
```

```
##                      ability   s.e.                team
## New England Patriots   1.769 0.1798 New England Patriots
## Pittsburgh Steelers    1.325 0.1733  Pittsburgh Steelers
## Indianapolis Colts     1.224 0.1711   Indianapolis Colts
## Green Bay Packers      1.104 0.1674    Green Bay Packers
## Philadelphia Eagles    1.075 0.1650  Philadelphia Eagles
## Denver Broncos         1.039 0.1687       Denver Broncos
```

Bradley-Terry model

Abilities plot

```
ggplot(data=abilities_brown, aes(x=reorder(team,ability), y=ability)) +
  geom_bar(stat="identity", fill=1:32) + coord_flip()+
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



Bradley-Terry model

Lets do predictions for a sample of games for Denver Broncos

```
broncos<-data.frame(team_home=rep("Denver Broncos", 4),
                    team_away=c("Cleveland Browns", "Tennessee Titans",
                                "Chicago Bears", "Detroit Lions"))
# we need to make team_home and team_away having the same factor structure
# as in the initial model

broncos$team_home<-factor(broncos$team_home,
                          levels(nfl3$team_away))

broncos$team_away<-factor(broncos$team_away,
                          levels(nfl3$team_away))
str(broncos)
```

```
## 'data.frame':    4 obs. of  2 variables:
##  $ team_home: Factor w/ 32 levels "Arizona Cardinals",..: 10 10 10 10
##  $ team_away: Factor w/ 32 levels "Arizona Cardinals",..: 8 31 6 11
```

```
broncos
```

```
##        team_home        team_away
## 1 Denver Broncos Cleveland Browns
## 2 Denver Broncos Tennessee Titans
## 3 Denver Broncos    Chicago Bears
## 4 Denver Broncos    Detroit Lions
```

Bradley-Terry model

```
#This is the probability of Denver Broncos winning

broncos_prob<-predict(model, newdata=broncos, level=2, type="response")
broncos_prob
```

```
##      1      2      3      4
## 0.7387 0.5833 0.5932 0.6994
```

Make a dataframe with probs

```
broncos_df<-data.frame(broncos, ht_w=broncos_prob,
                               at_w=1-broncos_prob)
broncos_df
```

```
##        team_home        team_away   ht_w   at_w
## 1 Denver Broncos Cleveland Browns 0.7387 0.2613
## 2 Denver Broncos Tennessee Titans 0.5833 0.4167
## 3 Denver Broncos    Chicago Bears 0.5932 0.4068
## 4 Denver Broncos    Detroit Lions 0.6994 0.3006
```

Bradley-Terry model

Probabilities are predicted using the following formula

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$

$P(\text{j beats i}) = \frac{e^{\alpha_j}}{e^{\alpha_i} + e^{\alpha_j}}$

Bradley-Terry model

Calculating probabilities manually for Denver Broncos-Cleveland Browns

```
#first get abbilities
broncos_ab <- abilities[rownames(abilities)=="Denver Broncos",1]
browns_ab <- abilities[rownames(abilities)=="Cleveland Browns",1]

broncos_ab
```

```
## [1] 0.6187
```

```
browns_ab
```

```
## [1] -0.4206
```

```
# probability of Broncos to win
exp(broncos_ab)/(exp(broncos_ab)+exp(browns_ab))
```

```
## [1] 0.7387
```

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i}+e^{\alpha_j}}$

Bradley-Terry model with order effect

Bradley-Terry model with order effect

Bradley-Terry model with order effect

- In certain types of application some or all contests have an associated 'bias', related to the order in which items are presented to a judge or with the location in which a contest takes place, for example.
- In sports this can be related to home advantage: the ability of the team is different when it plays at home compared to when it plays away.
- In this case the model has the following structure

z = 1 if i has the supposed advantage and z = -1 if j has it z = 0 if there is no advantage

Bradley-Terry model with order effect

We need to do several data transformations to get the model to account home advantage factor.

```
nfl3$team_home <- data.frame(team = nfl3$team_home, at.home = 1)
nfl3$team_away <- data.frame(team = nfl3$team_away, at.home = 0)
```

The result is a grouped_df

```
str(nfl3)
```

```
## grouped_df [992 x 4] (S3: grouped_df/tbl_df/tbl/data.frame)
##  $ team_home:'data.frame':   992 obs. of  2 variables:
##   ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 1 1 1 1 1 1 1 1 1 1 ...
##   ..$ at.home: num [1:992] 1 1 1 1 1 1 1 1 1 1 ...
##  $ team_away:'data.frame':   992 obs. of  2 variables:
##   ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 2 3 4 5 6 7 8 9 10 11 ...
##   ..$ at.home: num [1:992] 0 0 0 0 0 0 0 0 0 0 ...
##  $ ht       : num [1:992] 2 1 1 2 0 2 3 6 1 7 ...
##  $ at       : num [1:992] 1 2 1 4 3 0 0 2 3 1 ...
##  - attr(*, "groups")= tibble [32 x 2] (S3: tbl_df/tbl/data.frame)
##   ..$ team_home:'data.frame':    32 obs. of  2 variables:
##   .. ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 1 2 3 4 5 6 7 8 9 10 ...
##   .. ..$ at.home: num [1:32] 1 1 1 1 1 1 1 1 1 1 ...
##   ..$ .rows    : list<int> [1:32]
##   .. ..$ : int [1:31] 1 2 3 4 5 6 7 8 9 10 ...
##   .. ..$ : int [1:31] 32 33 34 35 36 37 38 39 40 41 ...
##   .. ..$ : int [1:31] 63 64 65 66 67 68 69 70 71 72 ...
##   .. ..$ : int [1:31] 94 95 96 97 98 99 100 101 102 103 ...
##   .. ..$ : int [1:31] 125 126 127 128 129 130 131 132 133 134 ...
##   .. ..$ : int [1:31] 156 157 158 159 160 161 162 163 164 165 ...
##   .. ..$ : int [1:31] 187 188 189 190 191 192 193 194 195 196 ...
##   .. ..$ : int [1:31] 218 219 220 221 222 223 224 225 226 227 ...
##   .. ..$ : int [1:31] 249 250 251 252 253 254 255 256 257 258 ...
##   .. ..$ : int [1:31] 280 281 282 283 284 285 286 287 288 289 ...
##   .. ..$ : int [1:31] 311 312 313 314 315 316 317 318 319 320 ...
##   .. ..$ : int [1:31] 342 343 344 345 346 347 348 349 350 351 ...
##   .. ..$ : int [1:31] 373 374 375 376 377 378 379 380 381 382 ...
```

Bradley-Terry model with order effect

```
head(nfl3$team_home, n=5)
```

```
##                 team at.home
## 1 Arizona Cardinals       1
## 2 Arizona Cardinals       1
## 3 Arizona Cardinals       1
## 4 Arizona Cardinals       1
## 5 Arizona Cardinals       1
```

Bradley-Terry model with order effect

```
model2 <- BTm(cbind(ht, at), team_home, team_away, formula=~team+at.home,
              data=nfl3, id="team")
```

```
model2$coefficients
```

```
##      teamAtlanta Falcons     teamBaltimore Ravens        teamBuffalo Bills
##                  0.37386                  0.62493                  0.05367
##    teamCarolina Panthers        teamChicago Bears   teamCincinnati Bengals
##                  0.31117                  0.24678                  0.24130
##     teamCleveland Browns       teamDallas Cowboys       teamDenver Broncos
##                 -0.42821                  0.42081                  0.63230
##        teamDetroit Lions    teamGreen Bay Packers       teamHouston Texans
##                 -0.23142                  0.69886                  0.09027
##   teamIndianapolis Colts teamJacksonville Jaguars   teamKansas City Chiefs
##                  0.82082                 -0.08398                  0.33933
## teamLos Angeles Chargers     teamLos Angeles Rams       teamMiami Dolphins
##                  0.37467                  0.04645                  0.29350
##    teamMinnesota Vikings teamNew England Patriots   teamNew Orleans Saints
##                  0.33000                  1.37599                  0.51851
##      teamNew York Giants        teamNew York Jets      teamOakland Raiders
##                  0.32672                  0.23093                 -0.11014
##  teamPhiladelphia Eagles  teamPittsburgh Steelers  teamSan Francisco 49ers
##                  0.66638                  0.92438                  0.05285
##     teamSeattle Seahawks teamTampa Bay Buccaneers     teamTennessee Titans
##                  0.48433                  0.04148                  0.28859
##  teamWashington Redskins                  at.home
##                  0.03614                  0.30285
```

Bradley-Terry model with order effect

Let make predictions for Broncos vs Browns by hand, assuming that Broncos are playing at home

```
db <- coefficients(model2)['teamDenver Broncos']
db
```

```
## teamDenver Broncos
##             0.6323
```

```
cb <- coefficients(model2)['teamCleveland Browns']
cb
```

```
## teamCleveland Browns
##              -0.4282
```

```
h <- coefficients(model2)['at.home']
h
```

```
## at.home
##  0.3028
```

Bradley-Terry model with order effect

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$

Probability of Broncos to win at home

```
w_h <- exp(db+h)/(exp(db+h)+exp(cb))
w_h
```

```
## teamDenver Broncos
##             0.7963
```

Odds of Broncos to win at home

```
od_h <- w_h/(1-w_h)
od_h
```

```
## teamDenver Broncos
##              3.909
```

Bradley-Terry model with order effect

Playing at "neutral" field, no advantage for any team

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$

```
w_n <- exp(db)/(exp(db)+exp(cb))
w_n
```

```
## teamDenver Broncos
##             0.7428
```

Odds of wining at "neutral" field

```
od_n <- w_n/(1-w_n)
od_n
```

```
## teamDenver Broncos
##              2.888
```

Bradley-Terry model with order effect

Odds ratio and home team advantage

```
od_h/od_n
```

```
## teamDenver Broncos
##              1.354
```

```
exp(coefficients(model2)['at.home'])
```

```
## at.home
##   1.354
```

$e^{\text{at.home}}$ is the odd multiplier for a home team to win compared to the neutral field

Bradley-Terry model with order effect

The same game, now Broncos is away, Browns is at home

$P(\text{j beats i}) = \frac{e^{\alpha_j + h}}{e^{\alpha_i} + e^{\alpha_j + h}}$

```
w_a <- exp(db-h)/(exp(db-h)+exp(cb))
w_a
```

```
## teamDenver Broncos
##             0.6808
```

Probability of Browns to win at home

```
1 - exp(db-h)/(exp(db-h)+exp(cb))
```

```
## teamDenver Broncos
##             0.3192
```

```
# OR
exp(cb+h)/(exp(db)+exp(cb+h))
```

```
## teamCleveland Browns
##               0.3192
```

Bradley-Terry model

Making predictions for the series of games

```
broncos <- data.frame(team_home=rep("Denver Broncos", 4),
                      team_away=c("Cleveland Browns", "Tennessee Titans",
                                  "Chicago Bears", "Detroit Lions"))

broncos$team_home<-data.frame(team = factor(broncos$team_home,
                                            levels(nfl3$team_home$team)),
                                            at.home = 1)
broncos$team_away<-data.frame(team = factor(broncos$team_away,
                                            levels(nfl3$team_home$team)),
                                            at.home = 0)
broncos
```

```
##   team_home.team team_home.at.home   team_away.team team_away.at.home
## 1 Denver Broncos                 1 Cleveland Browns                 0
## 2 Denver Broncos                 1 Tennessee Titans                 0
## 3 Denver Broncos                 1    Chicago Bears                 0
## 4 Denver Broncos                 1    Detroit Lions                 0
```

Bradley-Terry model

```
br2 <- predict(model2, newdata=broncos, level=2, type="response")
br2
```

```
##      1      2      3      4
## 0.7963 0.6562 0.6656 0.7625
```